LAB 02

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Data validation
TSC reference	A3, K5, K9
Duration	20 min
Simulator page	http://localhost:8000/#/data?tab=reference&ref=AAPL
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's traders watch DBS Group (SGX:D05) and Apple (NASDAQ:AAPL) on TradingView, while the pipeline trains and trades on data from an API. Before any model is built, the desk must prove that the pipeline's bars agree with what traders see on the chart - and understand why adjusted research prices differ from the unadjusted prices on the screen and on an order ticket.

Objective

Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 02. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 02: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Read TradingView's Data Window

In TradingView open NASDAQ:AAPL, daily timeframe, hover the bar of 2025-12-31 and read open/high/low/close in the Data Window (default settings: dividends NOT adjusted).

Open: http://localhost:8000/#/data?tab=reference&ref=AAPL (Data > Validate vs TradingView (AAPL))

Check: You have four numbers from TradingView.

Step 2. Compare with RAW and ADJUSTED bars

Type your TradingView values into the 'Your TradingView reading' row. The app shows the difference against RAW (split-adjusted only) and ADJUSTED (split + dividend) bars.

Open: http://localhost:8000/#/data?tab=reference&ref=AAPL (Data > Validate vs TradingView)

Check: Verdict 'PASS - TradingView matches RAW' (to within 0.1%).

Step 3. Repeat for SGX:D05 (DBS)

Select SGX:D05. DBS pays large dividends, so the adjustment factor is far from 1.

Open: http://localhost:8000/#/data?tab=reference&ref=D05.SI (Data > Validate vs TradingView (D05))

Check: You can explain why the ADJUSTED close is lower than the RAW close and why a model trained on adjusted prices must never be fed raw live prices.

Step 4. Inspect the chart

Open the AAPL chart and compare its shape with TradingView.

Open: http://localhost:8000/#/data?tab=chart&symbol=AAPL (Data > Chart AAPL)

Check: The candles match TradingView's daily chart (allowing for the adjustment).

Expected in the app: A PASS / FAIL validation for SGX:D05 and NASDAQ:AAPL on 2025-12-31, with the dividend-adjustment factor explained.

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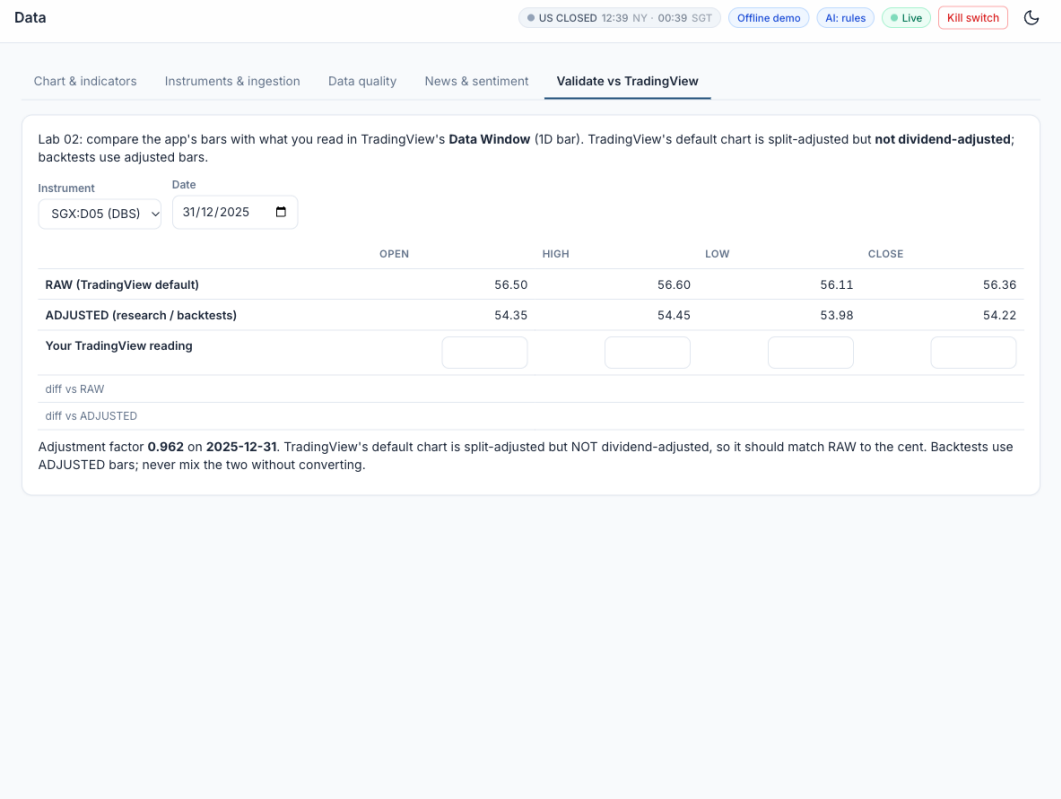


Figure: Lab 02 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.

Validate prices: adjusted versus unadjusted - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: download AAPL and D05.SI (DBS) daily bars for December 2025 twice - unadjusted (auto_adjust=False) and adjusted (auto_adjust=True). For 2025-12-31 print open, high, low and close from both, the ratio adjusted/unadjusted close (the dividend adjustment factor), and explain why TradingView's default chart matches the unadjusted prices.

Open in Claude | Open in ChatGPT | Open Google Colab

Guide: Sign Up for TradingView and Find the Tools You Will Use

The free TradingView plan is enough for Labs 2, 5 and 6. Do this before Lab 2.

1. Create the account

Go to tradingview.com and choose Sign up (top right). Register with email, Google or Apple, then confirm the activation email.

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2. Choose the free plan

Keep the free plan when asked. It shows adverts and limits the number of indicators per chart and the alert features; webhook alerts (Lab 17) need a paid plan.

3. Open Supercharts

From Products choose Supercharts. Search a symbol such as NASDAQ:AAPL or SGX:D05 in the symbol box at the top left and choose the daily (1D) timeframe.

4. Set the time zone

Click the time zone at the bottom right of the chart and choose (UTC+8) Singapore, so bar times match the course examples.

5. Locate the four panels the labs use

Indicators (top toolbar), the Pine Editor and the Strategy Tester (tabs at the bottom), and Trade / the Trading Panel (top right) for paper trading.

The free plan covers Labs 2, 5 and 6. Webhook alerts in Lab 17 need a paid plan - the lab includes a local simulator.

Debrief

Charting platforms are analytics tools (A3) and the API is the pipeline's data path (K5); a validation stage proves they agree before anything is automated. Knowing that adjusted and unadjusted prices are different forms of the same data (K9) prevents a classic error: training on adjusted history and then trading on raw quotes without converting.

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